

BUILT — Gap-closing optimization pass (v0.58.0)

Four gaps the speed passes hadn't touched: the OCR build, weak-PC startup, the live simulator, and the page/loupe round-trips. Each closed additively.

OCR build — dedup + adaptive DPI

- A density probe skips blank pages; identical rendered pages are md5-hashed and the OCR result is REUSED instead of re-inferred.
- Sparse pages render at a lower adaptive DPI (160 floor) — opt-in via --adaptive so accuracy is never silently traded.
- Batch log now reports dedup_reused. Corpus stays read-only (R1); applies on the next pass.

RESULT: Fewer GPU inferences per batch (dup pages reuse one result).

Legacy memory + cold-start warmup

- RPS now sizes the open-PDF LRU per mode (modern 8 / lite 3 / legacy 2) so a low-RAM PC keeps a small footprint.
- After rps_init the server warms the path (SELECT 1 + COUNT + one page) so the FIRST real request isn't the one paying for parse + cache fill.
- Per-mode SQLite cache/mmap unchanged; reversible.

RESULT: Smaller legacy footprint; first request no longer cold.

Circuit Lab MNA → Web Worker

- The continuous RUN loop's matrix solve now runs in circuitsim-worker.js, OFF the main thread; the page renders from posted snapshots so a heavy circuit can't stutter the UI.
- Edit / DC / single-Step stay inline + synchronous (unchanged). A WorkerSim shim exposes v()/i()/state() so draw() is untouched.
- Guaranteed inline fallback if Workers are absent or error (no breakage, R1).

RESULT: Run-loop math off the UI thread (validated; inline fallback).

⚡ Loupe + result hover-prefetch

- Hovering a search result warms its page render (debounced, de-duped) so the click opens from cache instantly.
- The loupe already upscales locally with zero latency; it now also prefetches the 4 neighbouring crisp crops so a slow drag stays sharp instead of flashing soft-then-sharp.
- Pure client-side; no new API surface.

RESULT: Click-to-open feels instant; loupe drag stays crisp.

NO OVERBUILD — EACH GAP CLOSED IN PLACE, FALLBACKS INTACT

None of these add a page or an API to learn. The OCR and startup wins are server-side and reversible; the Worker and prefetch wins are client-side with the original synchronous/inline paths kept as guaranteed fallbacks. Worker + shim validated in isolation (node --check + a mock-Worker run: init/dc/step posted, snapshots drawn, inline rebuilt on stop); hover-prefetch

dedup + URL verified. 23 pillars green: viewer_app imports clean.

BUILT diagram. Dark (R3). v0.58.0 · 2026-06-02 · viewer_ingest.py · rps.py/viewer_app.py warmup · circuitsim-worker.js + circuitlab.html · index.html. Additive (R1/R6).